Response to Reviewers

Training-Free Ranking from Pairwise Comparisons via Acyclic Graph Construction

Journal of Supercomputing — Major Revision

Soroush Vahidi

Cover note to the Editor

Dear Editor,

Thank you for arranging this careful review. I have prepared a major revision addressing all comments from Reviewers 1–4. The principal changes are:

- title revised to remove the word "Scalable";
- corrected add-back implementation using exact reachability-aware cycle safety (replacing the submitted fixed-topological proxy);
- secondary weighted min-cut structural exchange with proofs and regime-specific empirics;
- rebuilt baseline comparisons from a single canonical source and replaced headline tables with OURS-Reach (resolving the submitted Table 4/5 inconsistency and the topo-proxy/add-back mismatch);
- direct classical runtime comparisons (showing that OURS is slower than lightweight classical methods);
- comprehensive structural ablation, parameter sensitivity, timeout/coverage analysis, and family-aware statistics;
- clarified ranking–MWFAS theory and DF03 guarantee boundaries;
- manuscript-wide rewriting with a dedicated non-repetition pass.

The revised manuscript now includes the additional experiments and analyses requested by the reviewers. The point-by-point responses below cite the revised manuscript sections, propositions, tables, and figures.

Sincerely,
Soroush Vahidi

General reply to all reviewers

Dear Editor and Reviewers,

Thank you for the careful and constructive reviews. I agree that the original presentation did not sufficiently distinguish prior algorithmic components from the contributions of this work, left the Demetrescu–Finocchi guarantee ambiguous under practical timeouts, and did not provide the ablation, classical runtime, and statistical evidence needed for a major revision. The revised manuscript now separates prior local-ratio/add-back/refinement lineage, identifies exact reachability as an implementation-fidelity correction, positions the weighted min-cut exchange as a secondary new structural contribution, and clarifies the theoretical and empirical contributions.

Reviewer 1

Comment 1 (Positioning / novelty). *The contribution relative to prior feedback-arc-set / ranking work was not clearly delineated; the paper appeared to present the pipeline as more novel than supported.*

Response. I agree that the original positioning was insufficient. The Introduction was completely rewritten, Related Work was restructured (including an explicit relation to Vahidi & Koutis, arXiv:2412.16181), and a compact novelty-separation table was added. The revised manuscript no longer presents local-ratio cycle breaking, exact cycle-safe reinsertion, weight-ordered add-back, or Phase C refinement (adjacent-swap and order-preserving ternary) as newly invented. The title no longer contains "Scalable."

Changes in the manuscript: Sections 1 and 1.2; Table 1; title page.

Comment 2 (Ablation / sensitivity). *Provide ablations of Phase 1 only, Phase 1 with reinsertion, and the full pipeline; report upset metrics and runtime; study scale/density and sensitivity to zero tolerance, reinsertion passes, and refinement iterations.*

Response. A structural ablation with human-readable stage labels was added: A0 = Phase A only; A1 = Phase A + submitted topological-proxy add-back; A2 = Phase A + exact reachability add-back; A4 = A2 + Phase C refinement (adjacent-swap then order-preserving ternary); A5/A6 = optional min-cut on top of A2/A4. The principal empirical method is OURS-REACH = A4 (reachability + refinement; min-cut off). I have further revised Table 8 so that every within-panel stage comparison uses identical dataset support. Because the legacy fixed-topological/INS configuration (A1) and its refined full pipeline (A3) are available only on a smaller $n = 33$ scope, comparisons involving those variants are reported on the exact common subset shared with Phase A only (Panel (a): A0 $\rightarrow$ A1 $\rightarrow$ A3, $n = 33$), while the revised Phase A $\rightarrow$ exact-reachability-reinsertion $\rightarrow$ refinement progression is reported separately on its own larger $n = 77$ common-completion set (Panel (b): A0 $\rightarrow$ A2 $\rightarrow$ A4, OURS-REACH). Figure 2 was regenerated in the same matched-support panel layout (legacy $n = 33$; canonical $n = 77$); A5/A6 are omitted from that trajectory because they are secondary structural operators. This avoids conflating stage effects with differences in dataset composition. Both panels report all three requested upset measures and mean runtime. Directional significance for exact reachability is established by the paired tests in Table 7. Legacy multipass topo-proxy passes P2/P3 add little beyond P1; `zero_tol` is stable; refinement saturates quickly (R0 off; R1≈R2≈R3). Min-cut budgets K20/K50/K100 are reported in Section 3.8. Non-Finance A4 single-invocation algorithm runtimes are roughly 0.01–0.83 s (median ≈ 0.38 s) for $n \leq 602$; Finance is the large-dense boundary (Section 3.5). Table 8's runtime column uses this single-invocation algorithm quantity rather than the raw per-run harness timer.

Changes in the manuscript: Sections 3.1, 3.5–3.8; Tables 7–8; Figures 1–2.

Comment 3 (Algorithm readability). *The algorithm description needs complete, readable pseudocode and explicit parameters (ties, tolerances, timeouts).*

Response. A unified Algorithm 1 was inserted covering deterministic cycle search, residual updates, zero tolerance, forced-progress safeguards, wall-clock checks, exact reachability reinsertion, optional min-cut (gated on a DAG kept graph), Phase C adjacent-swap then order-preserving ternary refinement, and identity fallback. The pseudocode now flags that Proposition 2's DAG hypotheses apply only when Phase A exits acyclic; after a cyclic timeout exit the reachability scan may still run under the remaining budget, but min-cut is skipped. Table 2 lists canonical values including R0–R3, P1–P3, and K20/K50/K100.

Changes in the manuscript: Sections 2.3 and 2.8; Algorithm 1; Table 2.

Comment 4 (Theory / complexity / guarantees). *Clarify the ranking–MWFAS relationship, complexity claims, and whether the practical method inherits the Demetrescu–Finocchi approximation guarantee; provide an error discussion for fallback ordering.*

Response. Proposition 1 proves optimum-value equivalence $\mathrm{OPT}_{\mathrm{rank}} = \mathrm{OPT}_{\mathrm{MWFAS}}$ (many-to-many, not a bijection). For the audited Phase A implementation, the idealized full-completion complexity is $O(mn + m^2)$. Exact reachability add-back is stated as $O(m \log m + mn^2)$ for the dense suite implementation ($n \leq 4000$), with a sparse BFS/DFS fallback for larger n. Section 2.9 separates the idealized local-ratio guarantee from the practical finite-budget pipeline: the revised manuscript does *not* claim the DF03 guarantee for prematurely terminated runs. Proposition 4 shows that identity-order fallback admits no nontrivial general multiplicative bound.

Changes in the manuscript: Sections 2.2, 2.9–2.11; Propositions 1 and 4; Remark on DF03.

Comment 5 (Applications / future work). *Provide more concrete future directions and application context.*

Response. Dedicated Limitations and Future Work sections were added covering denser-graph theory, SCC/parallel engineering, candidate selection for min-cut, and application settings (preference aggregation, multi-source sensors, industrial ranking) as directions, not evaluated domains.

Changes in the manuscript: Sections 4–6.

Reviewer 2

Comment 1 (Novelty). *Clarify what is scientifically new relative to prior MWFAS / ranking work.*

Response. As in the reply to Reviewer 1 (Comment 1), the revised manuscript now separates prior lineage, fidelity correction, new theory, secondary min-cut, and expanded empirics (Table 1; Sections 1–2).

Comment 2 (Add-back contribution). *The submitted add-back / INS variants do not appear to contribute clearly; please diagnose and strengthen this stage.*

Response. I agree that the submitted fixed-topological proxy was too conservative and that multi-pass INS2/INS3 were not major quality innovations. Exact reachability restores the intended cycle-safety test and materially improves rankings in the ablation (A0→A2: 76/0/1; A1→A2: 32/0/1). The principal Tables 4–6 now report OURS-Reach (exact reachability + Phase C adjacent-swap and ternary refinement), not the submitted topo-proxy. INS multipass is retained only historically. As a secondary mechanism, a weighted min-cut exchange with monotone weighted-FAS improvement was added (Proposition 3; Section 3.9); min-cut is not part of the headline method. Structural A2→A5 / A4→A6 removed-weight gains remain valid because Phase C does not alter the retained graph; A6 upset deltas are not interpreted as "A4 scores plus min-cut."

Changes in the manuscript: Sections 2.5–2.6, 3.7–3.9; Algorithms 1–2; Table 7.

Comment 3 (GNN protocol / timing fairness). *Clarify what GNN runtimes include and whether hardware is comparable.*

Response. Archived GNN timings measure end-to-end training+evaluation, not inference-only latency. The exact archived GPU model ledger is unavailable; the manuscript therefore qualifies hardware comparability and does not claim matched stage-by-stage inference fairness.

Changes in the manuscript: Section 3.1; runtime discussion in Section 3.4.

Comment 4 (Timeout bias / missingness). *Timeouts and missing values may bias aggregates; please report coverage honestly.*

Response. Pairwise common-completion comparisons are now primary. Coverage is reported explicitly on the 78 loadable suite members (intended suite 80; two adjacency files unavailable): OURS-Reach 77/78 (Finance timeout); most classical methods 78/78; SyncRank 77/78 (Finance timeout); DIGRAC/ib 61/78 (primarily not-applicable rather than timeout). Finance is separated rather than averaged into quality macros. Missing values are not imputed as scores.

Changes in the manuscript: Sections 3.1, 3.4, 3.5.

Comment 5 (Oracle best-in-suite). *Best-in-suite / oracle envelopes should not be treated as deployable competitors.*

Response. I agree. Direct fixed-method comparisons are primary; oracle envelopes are de-emphasized and labeled as non-deployable post-hoc context only.

Changes in the manuscript: Section 3.1; captions of Tables 4–5.

Comment 6 (Statistics / dataset dependence). *Provide formal paired statistics and address Basketball-dominated dependence.*

Response. The revised manuscript reports W/T/L, median paired deltas, and—where exported—Wilcoxon tests with Holm adjustment and bootstrap CIs. Family-aware equal-family macros, leave-one-family-out, and Basketball-collapsed analyses are included. SpringRank comparisons are acknowledged as LOFO-fragile; BTL's and corrected RankCentrality's `upset_ratio` advantages persist. Ten classical methods were evaluated in the archived suite; Tables 4–6 display a prespecified principal subset of eight classical methods plus two GNN backbones (SVD-RS and eigenvector centrality omitted as near-duplicate spectral companions). An inherited GNNRank RankCentrality overwrite bug was corrected to the Negahban–Oh–Shah construction and only that baseline was recomputed; other baselines are unchanged.

Changes in the manuscript: Sections 3.1–3.3; Tables 4–6, 7.

Comment 7 (Deterministic repetitions). *Ten deterministic repetitions should not be interpreted as ranking robustness.*

Response. I agree. For deterministic non-GNN methods, repeats characterize runtime variability and timeout repeatability, not independent ranking observations.

Changes in the manuscript: Sections 3.1 and 4.

Comment 8 (Scalability). *The word "scalable" is not supported without qualification given dense/large cases.*

Response. "Scalable" was removed from the title. Empirical scope is qualified to the evaluated sparse and moderately dense graphs in the suite, with Finance as an explicit boundary.

Changes in the manuscript: Title; Sections 3.5–3.6, 4.

Comment 9 (Presentation / repetition). *The manuscript is repetitive and hard to navigate.*

Response. The experimental protocol was consolidated, Limitations were separated from Conclusion, and a manuscript-wide repetition audit was performed (novelty, DF03, INS, Finance, runtime accounting, contributions).

Changes in the manuscript: Throughout; Sections 3.1, 4–5.

Reviewer 3

Comment 1 (Ineffective add-back / INS1–3). *OURS-MFAS and INS variants appear nearly identical; add-back does not seem to help.*

Response. I agree with this diagnosis of the submitted manuscript: archived OURS-MFAS and INS3 were effectively duplicate topo-proxy configurations, and the fixed-topological rule was too conservative—so the submitted flagship add-back yielded no measurable ranking benefit. The revised Phase B uses exact reachability-aware reinsertion, and the principal comparison tables now evaluate OURS-Reach (reachability + Phase C adjacent-swap/ternary refinement) under the same GNNRank upset metrics as the baselines. Ablations show A2 improves the stage metric over Phase A on 76/77 datasets and over the legacy proxy on 32/33. Multipass INS2/INS3 are historical only; optional min-cut remains secondary (structural FAS-weight evidence; not a headline ranking claim).

Changes in the manuscript: Sections 2.5, 3.1, 3.7–3.8; Table 7.

Comment 2 (Table 4 / Table 5 inconsistency). *Submitted baseline tables were inconsistent (including classical values that did not match).*

Response. I agree and take responsibility for this inconsistency. The earlier aggregation procedure produced inconsistent summaries. I have regenerated the revised tables from a single canonical per-method export using consistent pairwise common-completion rules. For example, the canonical SpringRank median `upset_simple` in the verified full-suite export is 0.802724. Submitted unpaired Table 4/5 aggregates are not reused.

Changes in the manuscript: Section 3.1 (canonical source statement); Tables 4–5.

Comment 3 (Runtime versus classical methods). *Compare runtime directly against classical baselines, not only against GNNs.*

Response. Paired runtime W/T/L for OURS-Reach against classical methods is now reported. It is slower than most lightweight classical estimators (e.g., median ratios $\approx 5.7\times$ vs SpringRank and $\approx 170\times$ vs PageRank), comparable to SyncRank, and comparable to corrected RankCentrality ($\approx 1.03\times$ median paired ratio). The runtime advantage is relative to archived trained GNN end-to-end runs (about 37–45$\times$ faster on 60/60 common completions vs DIGRAC/ib), with the protocol qualification noted above.

Changes in the manuscript: Section 3.4; Table 6.

Comment 4 (Accuracy–runtime tradeoff / overclaiming). *Claims of accuracy and runtime advantage need to be metric-specific and honest.*

Response. Claims were revised to be metric-specific for OURS-REACH: strong simple/naive upset performance against the listed baselines; SpringRank remains family-sensitive despite a significant pairwise Holm test on `upset_simple`; BTL and corrected RankCentrality remain stronger on `upset_ratio`. An inherited GNNRank RankCentrality bug (transition matrix overwritten before the eigenvector step) was corrected and only that baseline recomputed. Ablations document why the submitted topo-proxy add-back failed and how exact reachability contributes. The revised manuscript does not claim universal dominance. Together with Comments 1–3, the three concrete requests—reconcile tables, adopt reachability-aware insertion in the flagship evaluation, and add classical runtime comparisons—are implemented.

Changes in the manuscript: Abstract; Sections 3.1–3.4, 5; Tables 4–6.

Reviewer 4

Comment 1 (Scientific novelty). *What exactly is scientifically new?*

Response. The Introduction and Related Work now answer this directly via prior/new separation and Table 1: formal ranking–MWFAS clarification and guarantee boundaries; fidelity restoration of exact add-back; secondary min-cut exchange; expanded empirical evaluation. Prior local-ratio/add-back/Phase C (adjacent-swap + ternary) components are credited, not reinvented.

Changes in the manuscript: Sections 1–2; Table 1.

Comment 2 (DF03 approximation guarantee). *Does the practical implementation inherit the Demetrescu–Finocchi guarantee?*

Response. The published guarantee applies to the idealized local-ratio procedure under its completion assumptions. The practical pipeline adds wall-clock termination, floating-point tolerances, forced-progress safeguards, and fallback output; therefore prematurely terminated practical runs do *not* automatically inherit the DF03 guarantee (Remark in Section 2.9). Algorithm 1 now branches so that Proposition 2's DAG hypotheses are not implied after a cyclic Phase A timeout, and min-cut is skipped unless the kept graph is a DAG. Proposition 4 addresses fallback error.

Changes in the manuscript: Sections 2.3, 2.9–2.10, 4; Algorithm 1.

Comment 3 (Multipass INS). *INS1/INS2/INS3 appear oversold.*

Response. Sensitivity analysis shows P1 restores many edges while P2/P3 add little ranking quality. Multipass INS variants are no longer framed as a core contribution.

Changes in the manuscript: Sections 2 (opening), 3.1, 3.8.

Comment 4 (GNN protocol). *Clarify GNN timing/accounting.*

Response. As in the reply to Reviewer 2 (Comment 3): end-to-end trained-GNN protocol, not inference-only; hardware ledger unavailable; claims qualified in Section 3.1.

Comment 5 ("Scalable"). *The title/text overstate scalability.*

Response. "Scalable" was removed from the title. Empirical behavior is described on the evaluated sparse and moderately dense graphs in the suite, with Finance as an explicit large-dense boundary.

Changes in the manuscript: Title; Sections 3.5–3.6, 4.

Comment 6 (Writing / repetition). *The writing is repetitive and sentences are too long.*

Response. The Introduction was rewritten, protocol and limitations were consolidated, duplicated novelty/DF03/INS/Finance explanations were removed, and long passages were shortened while preserving technical precision.

Changes in the manuscript: Throughout; Sections 1, 3.1, 4–5.

Comment 7 (MWFAS backbone / cycle-selection ablations). *Ablate the MWFAS backbone and related algorithmic choices.*

Response. The structural ablation addresses backbone stages (Phase A only; topo-proxy add-back; exact reachability; refinement; optional min-cut). Figure 2 now uses matched-support panels analogous to Table 8 (legacy A0→A1→A3 on $n = 33$; canonical A0→A2→A4 on $n = 77$) rather than a single A0–A6 median trajectory that mixed supports; A5/A6 are omitted from that figure and retained as secondary structural paired tests in Table 7. Cycle-selection sensitivity (C0/C1) shows a small but detectable effect on the A4 path ($|\mathrm{med}\,\Delta| \approx 10^{-3}$); it is reported without overstatement.

Changes in the manuscript: Sections 3.7–3.8; Table 7; Figure 2.

Closing

I thank the reviewers again for comments that substantially improved the clarity, correctness, and evidence standards of the manuscript. I hope the revised version addresses all concerns.